

# SMART EXPENSE TRACKER

Python-Based Expense Tracking and Spending Analysis

## Project Report

Technologies: Python • Pandas • Matplotlib • CSV • Google Colab

## 1. Introduction

Smart Expense Tracker is a beginner-friendly data analysis project developed using Python. It records expense categories and amounts, calculates total spending, performs category-wise analysis, identifies the highest spending category, and presents the results through visual charts.

## 2. Objective

- To record and organize daily expense information.
- To calculate total expenses using Pandas.
- To analyze spending category-wise.
- To identify the category with the highest spending.
- To visualize spending patterns using bar and pie charts.

## 3. Technologies Used

| Technology   | Purpose                                |
|--------------|----------------------------------------|
| Python       | Programming and project implementation |
| Pandas       | Data processing and analysis           |
| Matplotlib   | Data visualization                     |
| CSV          | Storing expense data                   |
| Google Colab | Development and execution environment  |

## 4. Key Features

- Expense data entry and storage.
- CSV file generation and reading.
- Total expense calculation.
- Category-wise expense aggregation.
- Highest spending category detection.
- Bar chart visualization.
- Pie chart expense distribution.

## 5. Data Analysis Process

The project stores expense records in a CSV file and loads the data into a Pandas DataFrame. The Amount column is summed to calculate total spending. GroupBy is used to aggregate spending by Category. The idxmax and max operations are then used to identify the highest spending category and its amount.

## 6. Visualization

A bar chart is used to compare spending amounts across categories. A pie chart is used to show the percentage distribution of expenses among the categories. These visualizations make spending patterns easier to understand.

## 7. Sample Project Data

| Category  | Amount (■) |
|-----------|------------|
| Food      | 200        |
| Travel    | 100        |
| Shopping  | 500        |
| Education | 300        |
| Food      | 150        |

## 8. Conclusion

The Smart Expense Tracker demonstrates how Python, Pandas, CSV files, and Matplotlib can be combined to perform a practical data analysis task. The project provides expense summaries and visual insights that can help users understand their spending patterns.

## 9. Future Enhancements

- Add monthly and yearly expense tracking.
- Add a user-friendly graphical interface.
- Add budget limits and alerts.
- Store data in a database.
- Create an interactive dashboard.